

Karnatak University, Dharwad

B.Sc. III Semester (Revised-NEP) – 2025-26

Mathematics Laboratory Manual

Paper - Practical on Algebra-II and Real Analysis-I

Program No. 1:

/*Construct Cayley table and test abelian for given finite set*/

kill(all)\$

S:{0,1,2}\$

bo(a,b):=mod(a+b,3)\$

flag:1\$

print("Cayley table")\$

print(true,"+|0 1 2")\$

print(true,"__\$")\$

for a in S do(

 printf(true,"~4d|",a),

 for b in S do(

 printf(true,"~4d",bo(a,b)),

 if bo(a,b)#bo(b,a) then flag:0),

 print(true,"~%")\$

if flag=1 then

print("Commutative axiom is satisfied")

else

print("Commutative axiom is not satisfied")\$

Output:

Cayley table

true+|0 1 2

 true__

0| 0 1 2

1| 1 2 0

2| 2 0 1

Commutative axiom is satisfied

Program No. 2:

/*To find generators & corresponding possible subgroups of G*/

```
kill(all)$
G:{1,-1,%i,-%i};
flag:0$
n:cardinality(G)$
m:0$
for a in G do(
  S:{},
  for j:1 thru n do(
    S:adjoin(a^j,S)),
  if G=S then(
    print(a,"is a generator"),
    m:m+1,flag:1)
  else
    print(a,"is not a generator"),
    print("Subgroup generated by",a,"is",S))$
if flag=0 then
print("G has no generator and not cyclic group ")
else
print("It is cyclic group and number of generators",m)$
```

Output:

```
      {-1,1,-%i,%i}
-1 is not a generator
Subgroup generated by -1 is {-1,1}
  1 is not a generator
Subgroup generated by 1 is {1}
  -%i is a generator
Subgroup generated by -%i is {-1,1,-%i,%i}
    %i is a generator
Subgroup generated by %i is {-1,1,-%i,%i}
It is cyclic group and number of generators 2
```

Program No. 3:

```
/*Verification of Lagrange's theorem*/
kill(all)$
G:set(1,-1,%i,-%i);
H:set(1,-1);
m:length(G);
n:length(H);
H1:cartesian_product(H,H)$
H2:makeset(a*b^-1, [a,b],H1)$
if(H2=H and mod(m, n)=0) then
disp("Lagrange's theorem is satisfied")
else
disp("Lagrange's theorem is not satisfied");
```

Output 1:

```
for H=(1,-1)
{-1,1,-%i,%i}
{1,i}
4
2
Lagrange's theorem is satisfied
```

Output 2:

```
for H=(1,i)
{-1,1,-%i,%i}
{1,i}
4
2
Lagrange's theorem is not satisfied
END
```

Program No. 4:

```
/* To check a set of real number attains supremum or infimum*/
kill(all)$
f(n):= n/(n+3);
A:makelist(f(n),n,1000)$
p:apply(min,A)$
q:apply(max,A)$
lim: limit(f(n),n,inf)$
if(lim<p) then
(
  if (lim#minf) then
    print("Infimum of the given function is", lim)
  else
    print("Function doesn't attain infimum")
)
else
print("Infimum of a given function is",p)$
if(lim>q) then
(
  if (lim#inf) then
    print("Supremum of the given function is", lim)
  else
    print("Function doesn't attain supremum")
)
else
print("Supremum of a given function is",q)$
```

Output 1:

$$f(n) := \frac{n}{n+3}$$

Infimum of a given function is $\frac{1}{4}$
Supremum of the given function is 1

Output 2:

$$f(n) := n$$

Infimum of a given function is 1
Function doesn't attain supremum

Output 3:

$$f(n) := \frac{1}{n}$$

Infimum of the given function is 0
Supremum of a given function is 1

Program No. 5(a):

```
/*Cauchy's First Theorem*/
kill(all)$
expr: read("Enter sequence term a_n terms of n:")$
n: read("Enter n:")$
x_n: sum(expr,k,1,n)$
L1: limit(expr, n, inf)$
L2: limit(x_n, n, inf)$
print("Limit of a_n=", L1)$
print("Limit of (a1+...+an)/n=", L2)$
if L1#ind and L2#ind and L1=L2 then
    print("Cauchy's first theorem verified")
else
    print("Theorem not satisfied")$
```

Output 1:

```
Enter sequence term a_n terms of n:1/n
Enter n:n
Limit of a_n= 0
Limit of (a1+...+an)/n= 0
Cauchy's first theorem verified
```

Output 2:

```
Enter sequence term a_n terms of n:(n+1)/n
Enter n:n
Limit of a_n= 1
Limit of (a1+...+an)/n= 1
Cauchy's first theorem verified
```

Output 3:

```
"Enter sequence term a_n terms of n: (-1)^n
Enter n: n
Limit of a_n = ind
Limit of (a1+...+an)/n = ind
Theorem not satisfied
```

Program No. 5(b):

```
/*Cauchy's Second Theorem*/  
kill(all)$  
expr:read("Enter sequence term a_n terms of n:")$  
L1:limit(expr^(1/n),n,inf)$  
L2:limit(subst(n+1,n,expr)/expr,n,inf)$  
print("Limit of (a_n)^(1/n)=",L1)$  
print("Limit of a_(n+1)/a_n=",L2)$  
if L1=L2 then  
    print("Cauchy's second theorem verified")  
else  
    print("Theorem not satisfied")$
```

Output 1:

```
Enter sequence term a_n terms of n: n!  
Limit of (a_n)^(1/n)= inf  
Limit of a_(n+1)/a_n= inf  
Cauchy's second theorem verified
```

Output 2:

```
Enter sequence term a_n terms of n: n!/n^n  
Limit of (a_n)^(1/n) = %e^-1  
Limit of a_(n+1)/a_n = %e^-1  
Cauchy's second theorem verified
```

Program No. 6:

```
/*Program to test geometric series*/
kill(all)$
a: read("Enter first term a:")$
r: read("Enter common ratio r:")$
print("Series:",a,",",a*r,",",a*r^2,",",a*r^3,",.....")$
if abs(r)<1 then
(
  print("Series converges"),
  print("Sum=",a/(1-r))
)
elseif (r>=1) then
  print("Series diverges")
elseif (r=-1) then
  print("Series oscillates finitely")
else
  print("Series oscillates infinitely")$
```

Output 1:

```
Enter first term a:3
Enter common ratio r:2
Series: 3 , 6 , 12 , 24 ,.....
Series diverges
```

Output 2:

```
Enter first term a:2
Enter common ratio r:1/2
Series: 2 , 1 ,  $\frac{1}{2}$  ,  $\frac{1}{4}$  ,.....
Series converges
Sum= 4
```

Output 3:

```
Enter first term a: 1
Enter common ratio r: -1
Series: 1, -1, 1, -1,.....
Series oscillates finitely
```

Output 4:

```
Enter first term a: 1
Enter common ratio r: -5
Series: 1, -5, 25, -125,.....
Series oscillates infinitely
```

Program No. 7:

```
/*Program to test p-series convergence*/
kill(all)$
p:read("Enter the value of p:")$
print("Series:  $\sum(1/n^p)$ ,  $n=1\dots$ , with  $p=$ ",p)$
if p>1 then
(
  print("Result: The series converges")
)
else
(
  print("Result: The series diverges")
)$
```

Output 1:

```
Enter the value of p: 2
Series:  $\sum(1/n^p)$ ,  $n=1\dots$ , with  $p= 2$ 
Result: The series converges
```

Output 2:

```
Enter the value of p: 1/2
Series:  $\sum(1/n^p)$ ,  $n=1\dots$ , with  $p= \frac{1}{2}$ 
Result: The series diverges
```

Program No. 8:

```
/*D' Alembert's Ratio Test*/
kill(all)$
print("D'Alembert's ratio test:")$
U(n):=n!/(n^n)$
ratio:U(n)/U(n+1)$
ss:sum(U(n),n,1,inf)$
print("Sum of the series is",ss)$
Lt:limit(ratio,n,inf)$
print("The limit value of U(n)/U(n+1) is",Lt)$
if Lt>1 then
print("The series is convergent")
else if Lt<1 then
print("The series is divergent")
else
print("The D' Alembert's ratio test fails")$
```

Output 1:

D'Alembert's ratio test:

Sum of the series is $\sum_{n=1}^{\infty} \frac{n!}{n^n}$

The limit value of $U(n)/U(n+1)$ is %e

The series is convergent

Output 2:

D'Alembert's ratio test :

Sum of the series is $\frac{\sum_{n=1}^{\infty} \frac{n!}{n^n}}{2}$

The limit value of $U(n)/U(n+1)$ is 0

The series is divergent

Program No. 9:

```
/*Raabe's Test*/
kill(all)$
print("Raabe's test:")$
U(n):=1/n^2$
ratio:n*(U(n)/U(n+1)-1)$
ss:sum(U(n),n,1,inf)$
print("Sum of the series is",ss)$
Lt:limit(ratio,n,inf)$
print("The limit value of n*(U(n)/U(n+1)-1) is",Lt)$
if Lt>1 then
print("The series is convergent")
else if Lt<1 then
print("The series is divergent")
else
print("The Raabe's test fails")$
```

Output 1:

Raabe's test:

Sum of the series is $\sum_{n=1}^{\infty} \frac{1}{n^2}$

The limit value of $n \cdot (U(n)/U(n+1)-1)$ is 2

The series is convergent

Output 2:

Raabe's test:

Sum of the series is $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$

The limit value of $n \cdot (U(n)/U(n+1)-1)$ is 1/2

The series is divergent

Program No. 10:

```
/*Cauchy's Root Test*/
kill(all)$
print("Cauchy's root test:")$
U(n):=(1/(n+1))^n$
nroot:U(n)^(1/n)$
ss:sum(U(n),n,1,inf)$
print("Sum of the series is",ss)$
Lt:limit(nroot,n,inf)$
print("The limit value of U(n)^(1/n) is",Lt)$
if Lt<1 then
print("The series is convergent")
else if Lt>1 then
print("The series is divergent")
else
print("The Cauchy's root test fails")$
```

Output 1:

Cauchy's root test:

Sum of the series is $\sum_{n=1}^{\infty} \frac{1}{(n+1)^n}$

The limit value of $U(n)^{(1/n)}$ is 0

The series is convergent

Output 2:

Cauchy's root test:

Sum of the series is $\sum_{n=1}^{\infty} \frac{n^{(n^2)}}{(n-1)^{(n^2)}}$

The limit value of $U(n)^{(1/n)}$ is %e

The series is divergent

Program No. 11:

```
/*Verification of Cauchy's integral test*/
kill(all)$
print("Cauchy's Integral Test:")$
U(n):=1/(n^2+1)$
ss:=sum(U(n),n,1,inf)$
print("U(n)=",ss)$
f(x):=1/(x^2+1)$
d(x):=ratsimp(f(x)-f(x+1))$
print("f(x)-f(x+1):",d(x))$
if d(1)> 0 then
(
  print("Sequence is monotonically decreasing"),
  print("Cauchy's integral test applicable"),
  assume(n>1),
  S:=integrate(f(x),x,1,n),
  L:=limit(S,n,inf),
  if L≠inf then
  (
    print("f(x) is convergent"),
    print("Here by Cauchy's integral test U(n) is convergent")
  )
else
  (
    print("f(x) is divergent"),
    print("Hence, by Cauchy integral test, U(n) is divergent")
  )
)
else
  print("Cauchy's integral test is not applicable")$
```

Output 1:

Cauchy's Integral Test:

$$U(n) = \sum_{n=1}^{\infty} \frac{1}{n^2 + 1}$$

$$f(x) - f(x+1): \frac{2x+1}{x^4 + 2x^3 + 3x^2 + 2x + 2}$$

Sequence is monotonically decreasing

Cauchy's integral test applicable

f(x) is convergent

Here by Cauchy's integral test U(n) is convergent

Output 2:

Cauchy's Integral Test:

$$U(n) = \sum_{n=1}^{\infty} \frac{1}{2n+3}$$

$$f(x) - f(x+1): \frac{2}{4x^2 + 16x + 15}$$

Sequence is monotonically decreasing

Cauchy's integral test applicable

$f(x)$ is divergent

Hence, by Cauchy integral test, $U(n)$ is divergent

Program No. 12:

```
/*Example on alternating series using Leibnitz theorem*/
kill(all)$
U(n):=1/n^2$
gseries:sum((-1)^(n-1)*U(n),n,1,inf)$
print("The given series is:",gseries)$
d(n):=ratsimp(U(n)-U(n+1))$
Lt:limit(U(n),n,inf)$
print("U(n)-U(n+1):",d(n))$
if d(1)>0 then
(
  print("Here Un is decreasing sequence"),
  print("Leibnitz first condition is satisfied"),
  if Lt=0 then
  (
    print("Limit of Un=0"),
    print("Leibnitz second condition is satisfied"),
    print("The given series is convergent")
  )
else
  (
    print("Leibnitz second condition is not satisfied"),
    print("Leibnitz test is not applicable")
  )
)
else
  print("Un is increasing sequence and hence Leibnitz test is not applicable")$
```

Output 1:

The given series is: $\sum \frac{(-1)^{n-1}}{n^2}$
U(n)-U(n+1): $\frac{2n+1}{n^4+2n^3+n^2}$
Here Un is decreasing sequence
Leibnitz first condition is satisfied
Limit of Un=0
Leibnitz second condition is satisfied
The given series is convergent

Output 2:

The given series is: $\sum_{n=1}^{\infty} \frac{n(-1)^{n-1}}{n+1}$

$$U(n) - U(n+1): -\left(\frac{1}{n^2 + 3n + 2}\right)$$

U_n is increasing sequence and hence Leibnitz test is not applicable